

● EASY

● ALGEBRA

Linear equations in one variable

03

10 questions · ~15 min

Q1

ALGEBRA

What value of x is the solution to the equation $16x + 24 = 24x$?

A. -4

B. $\frac{3}{10}$

C. $\frac{1}{3}$

D. 3

$$(p + 3) + 8 = 10$$

What value of p is the solution to the given equation?

- A. -1
- B. 5
- C. 15
- D. 21

$$10 = 2x + 4$$

How many solutions exist to the equation shown above?

- A. None
- B. Exactly 1
- C. Exactly 3
- D. Infinitely many

Q4

GRID-IN

ALGEBRA

$$2.6 + x = 2.8$$

What value of x is the solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A manager is responsible for ordering supplies for a shaved ice shop. The shop's inventory starts with 4,500 paper cups, and the manager estimates that 70 of these paper cups are used each day. Based on this estimate, in how many days will the supply of paper cups reach 1,700?

- A. 20
- B. 40
- C. 60
- D. 80

If $3x - 27 = 24$, what is the value of $x - 9$?

- A. 1
- B. 8
- C. 24
- D. 35

$$3x + 5(x + 4) = 76$$

What value of x is the solution to the given equation?

- A. 7
- B. 8
- C. 56
- D. 72

$$3x + 21 = 3x + k$$

In the given equation, k is a constant. The equation has infinitely many solutions. What is the value of k ?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$8x = 88$$

What value of x is the solution to the given equation?

- A. 11
- B. 80
- C. 96
- D. 704

John paid a total of \$ 165 for a microscope by making a down payment of \$ 37 plus p monthly payments of \$ 16 each. Which of the following equations represents this situation?

A. $16p - 37 = 165$

B. $37p - 16 = 165$

C. $16p + 37 = 165$

D. $37p + 16 = 165$